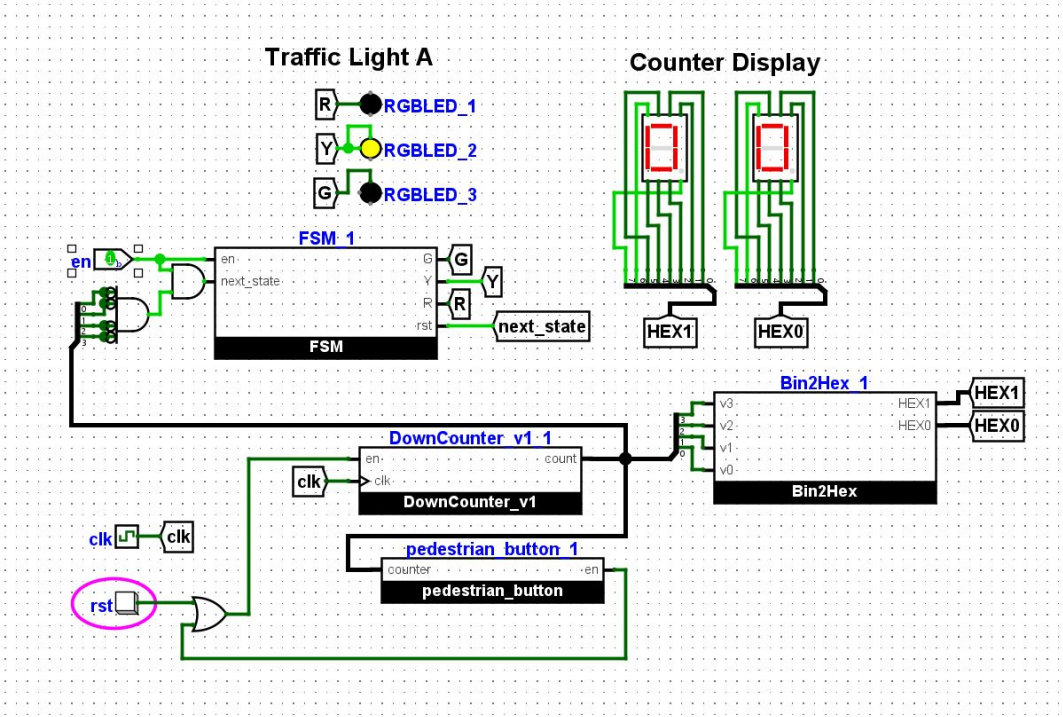
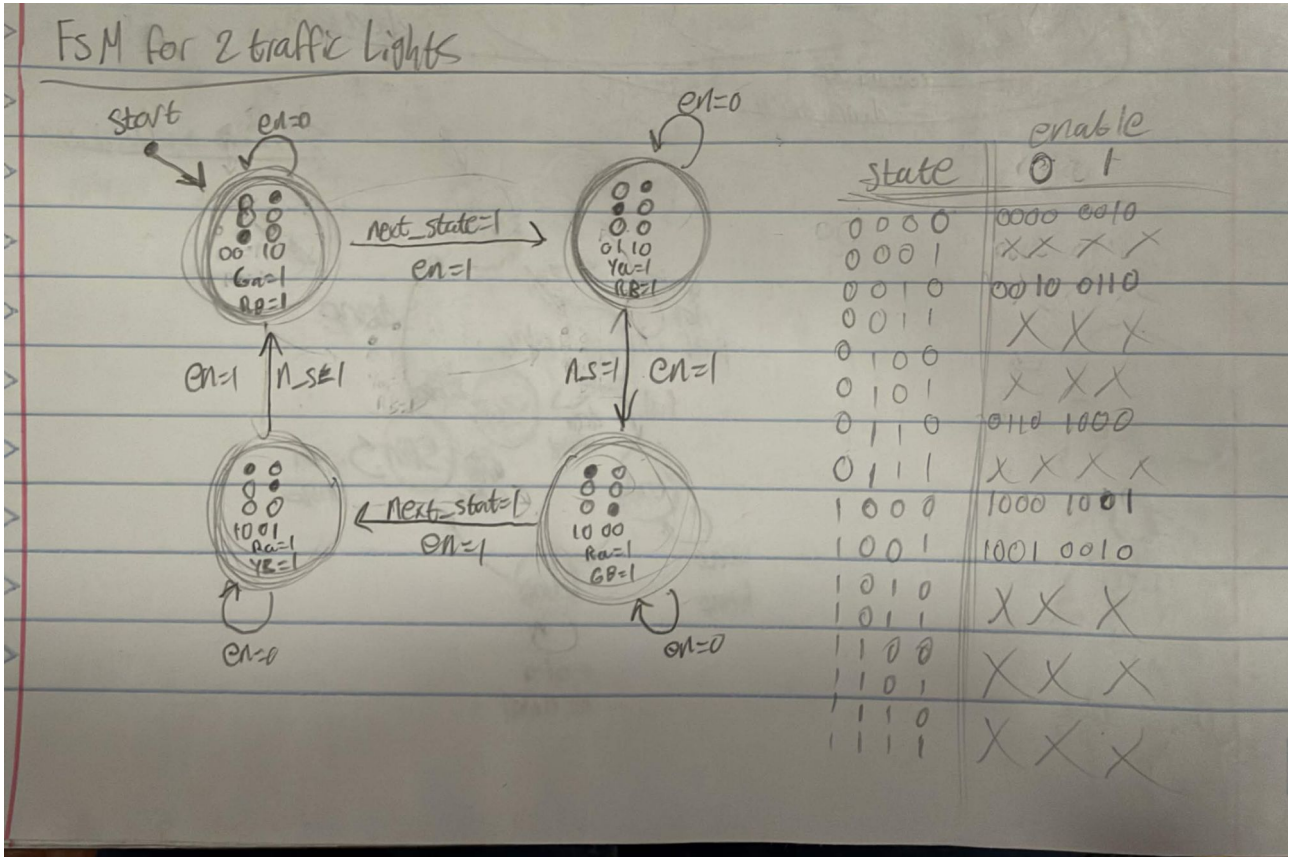
traffic light setup



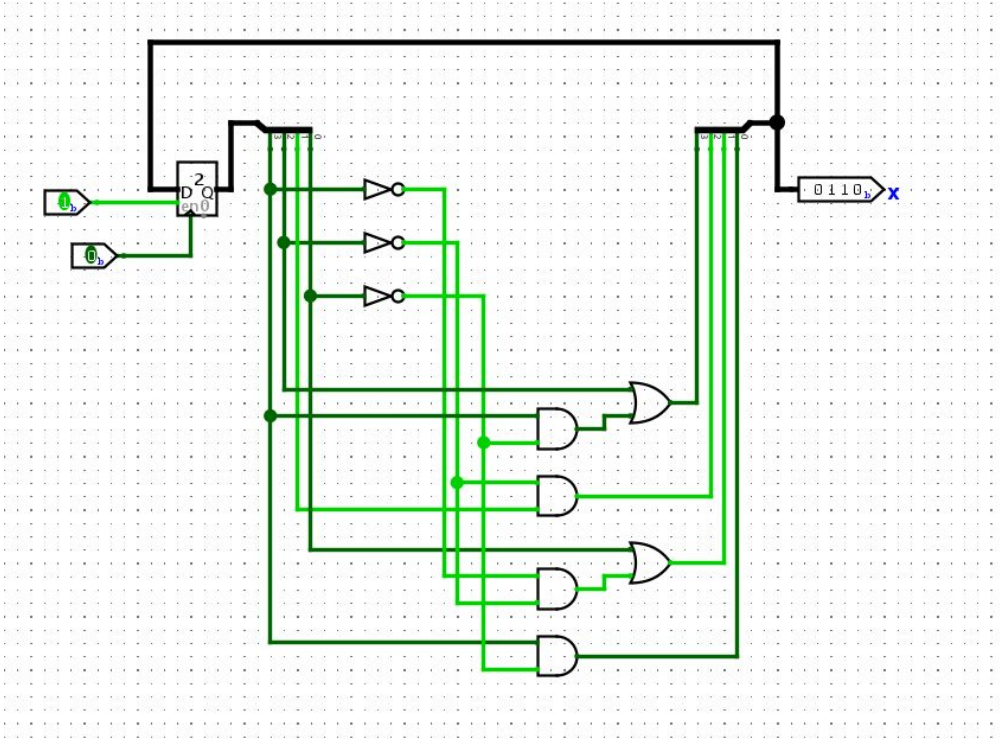
taken from previous lab

fsm

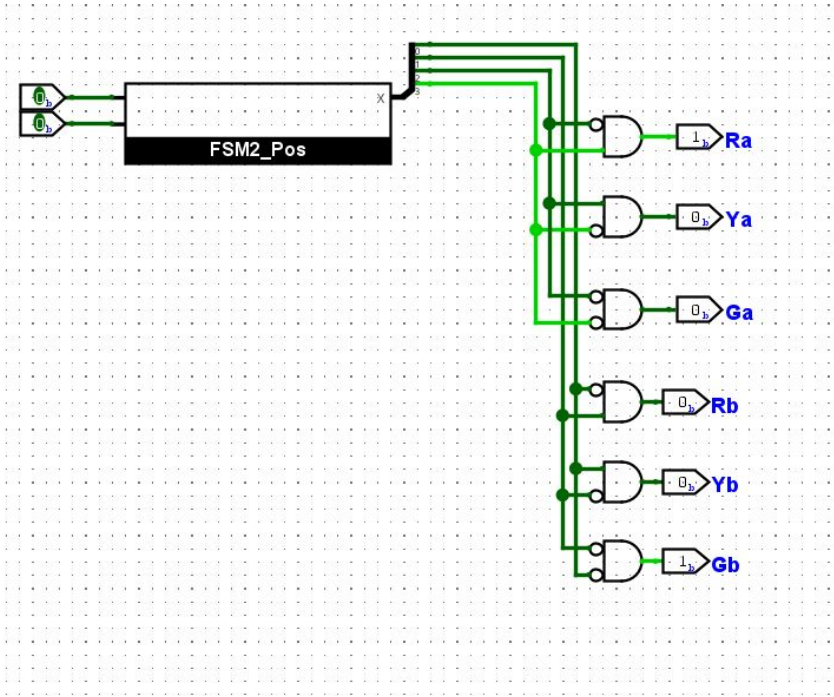


a bit crude on the truth table but it works

fsm implementation

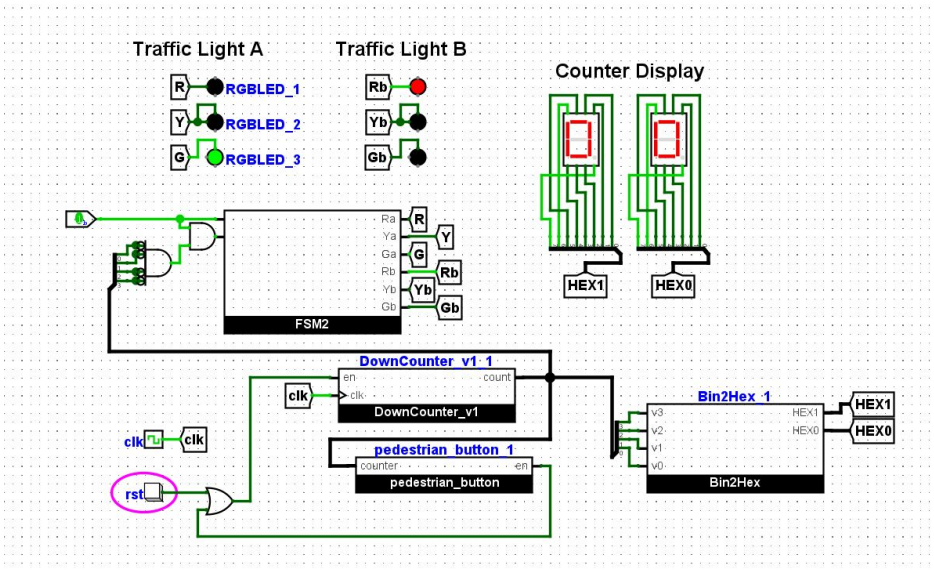


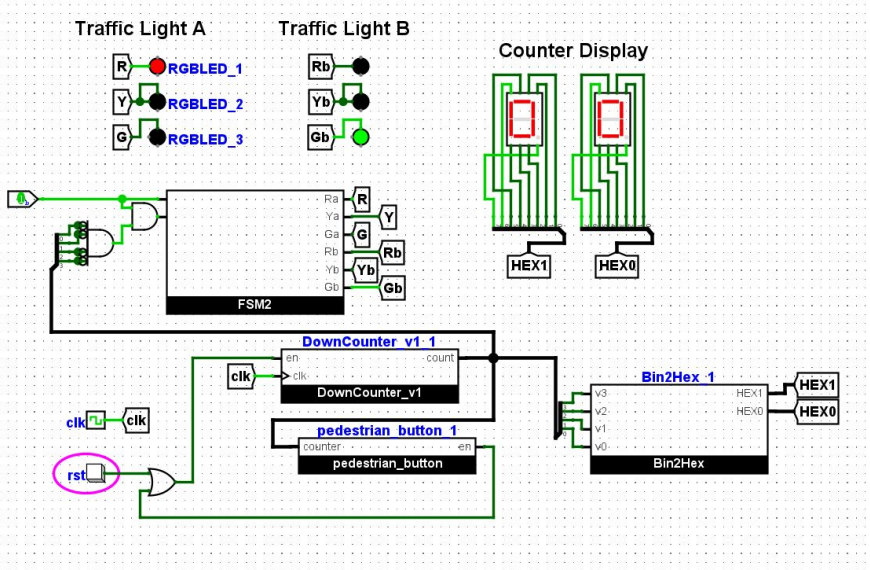
pos for truth table



cycles thru all 4 states

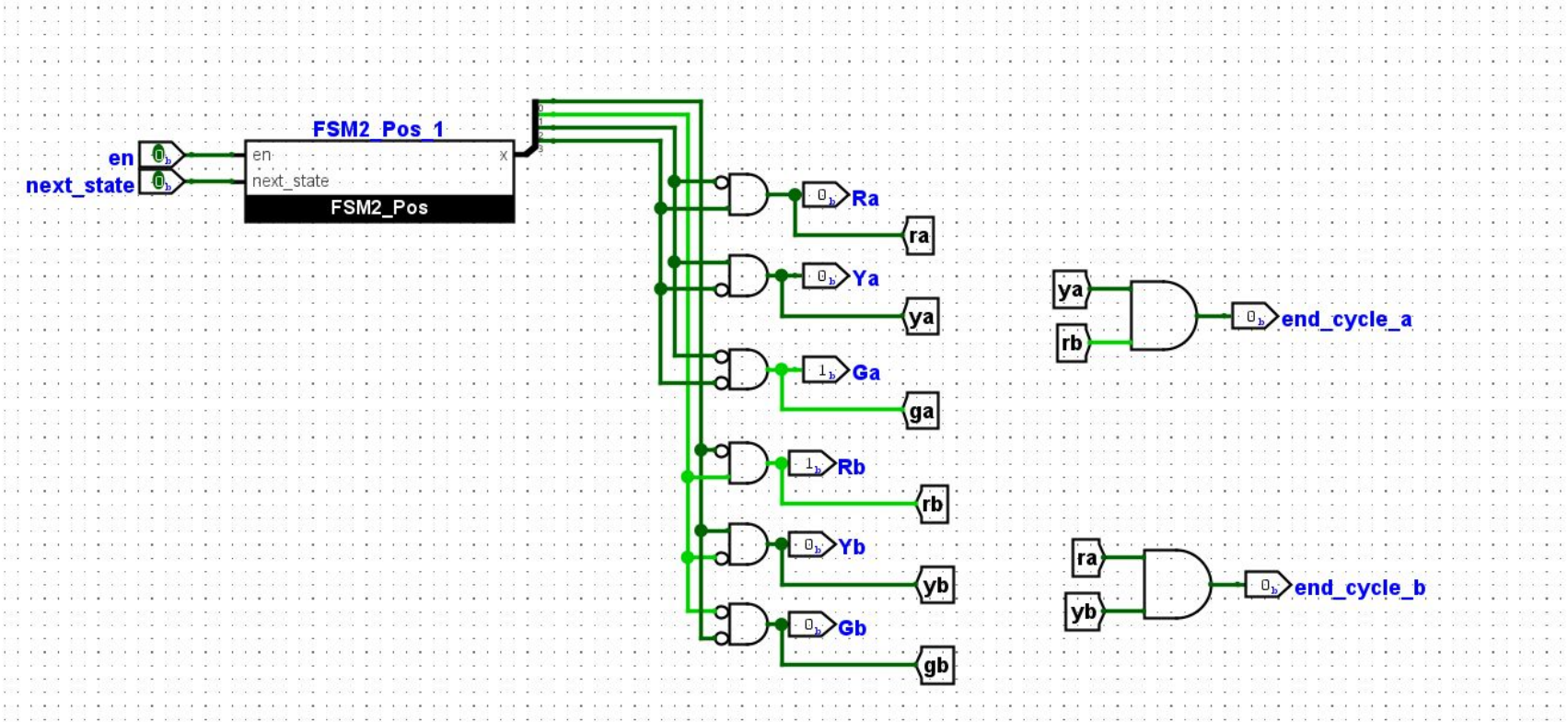
integration with lights





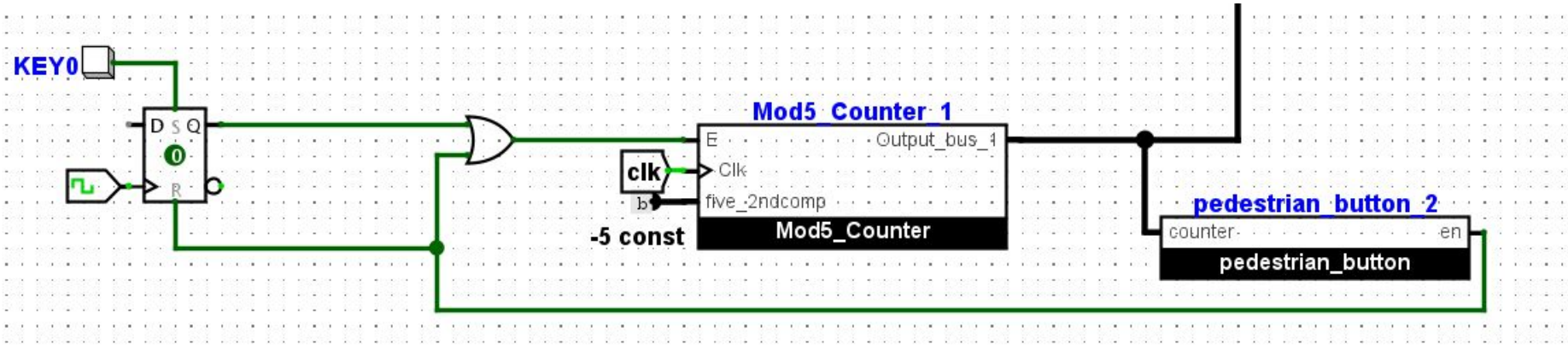
starts at green with light b red, and light b turns green when light a turns red. there is no overlap between the two

fsm revison

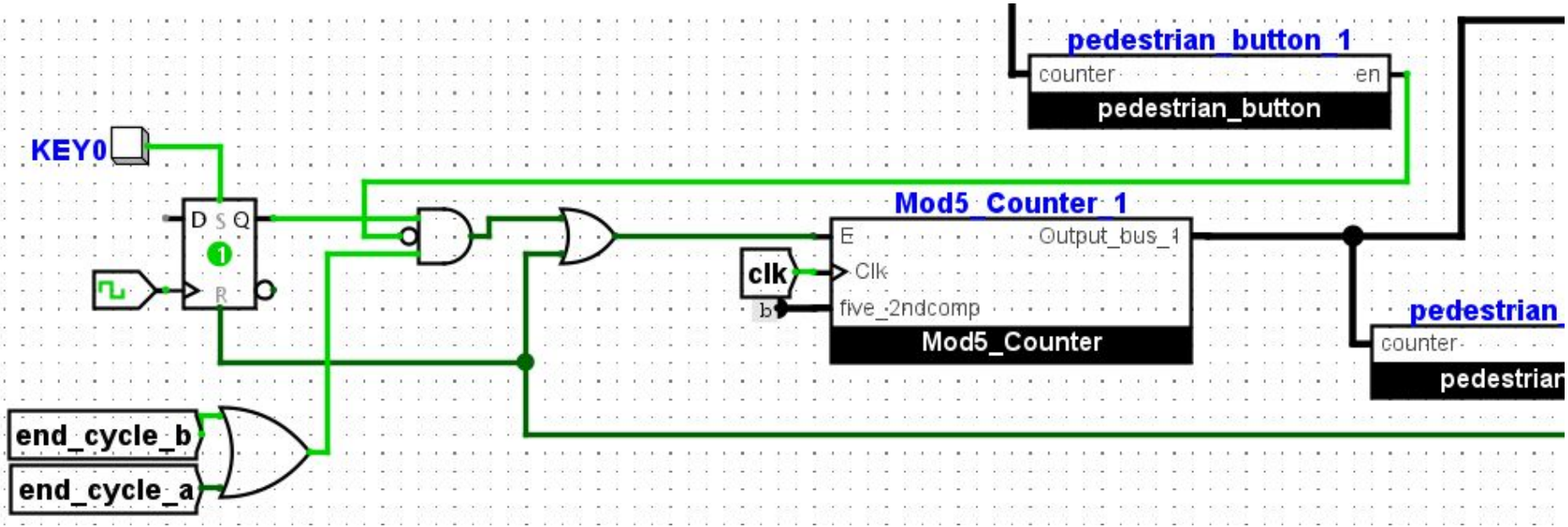


added 2 outputs that denote “end of cycle” states

10-0 counter

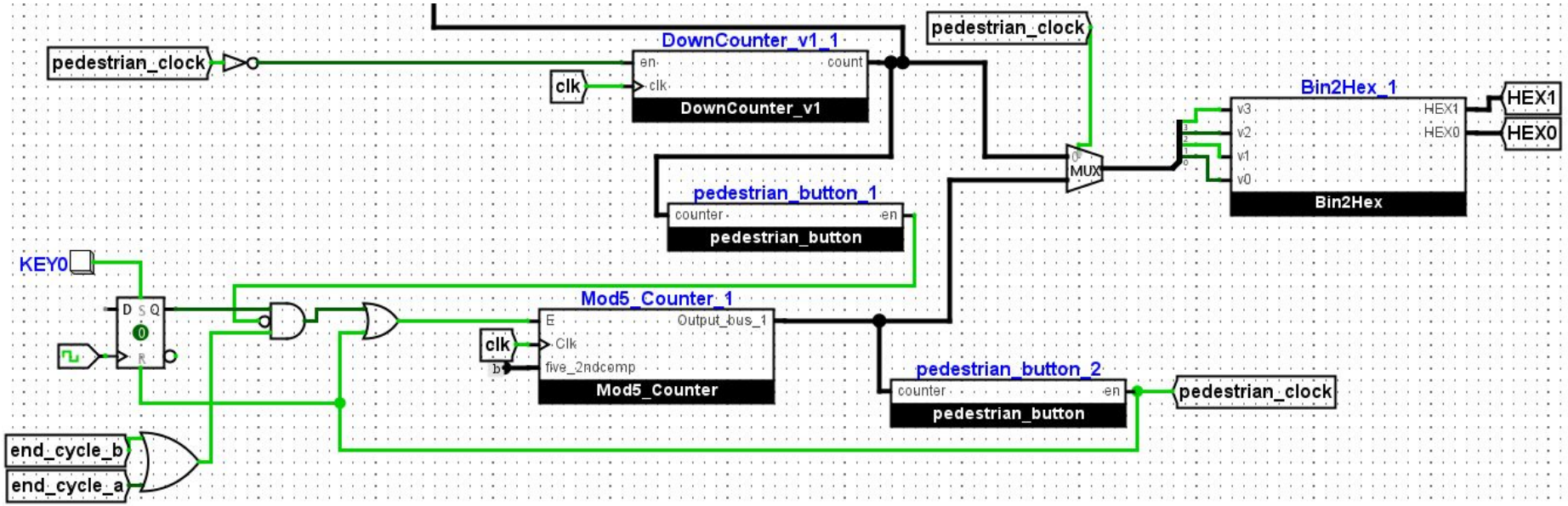


saves the button press until a certain condition is met (pending), counts from 10-0 and resets the button press



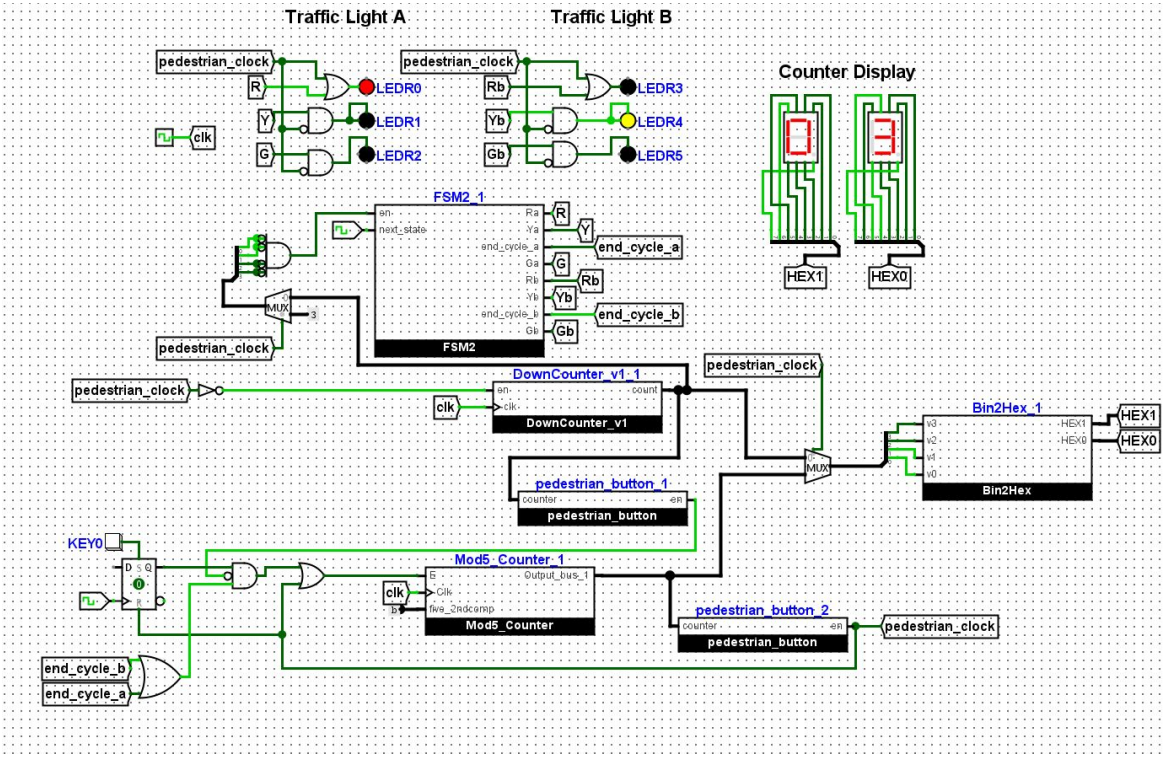
now only runs when the 3-0 count reaches 0, the button press has been saved AND the lights are at one of their end cycles. nice!

combination with 3-0 counter



using the output from pedestrian_button_2 (used thru the tunnel pedestrian_clock) that detects when the 10-0 counter is on, this logic successfully displays 3-0 or 10-0 with no overlap or janky timing

final implimentation



added: override for red LEDs so they both show red when pedestrian_clock is on, new logic that feeds into the fsm that makes each color swap between 0 and 3 as intended